## Business Brief: AI-Powered Call Routing for CBRE Facilities Management

### Problem Statement

CBRE's automated call center system, built to triage building maintenance requests, is critically unreliable in its urgency classification. The most prevalent symptom: low-severity issues like broken doors and minor leaks are being routed to emergency services (911). No human oversight exists to catch misroutes before execution, driving up costs and delaying legitimate resolutions.

### Proposed Solution

An AI-driven routing system that replaces the flawed classification layer, based on CBRE's historical call data and integrated directly with their existing ticketing system.

**Core pipeline:**

1. **Entity Extraction** — Parse call transcripts for issue type, location, and severity indicators
2. **Classification** — Map entities to predefined issue codes and urgency tiers using historical routing outcomes as context labels
3. **Routing** — Dispatch to the correct team based on learned patterns
4. **Human Review Gate** — Flag low-confidence classifications before execution, eliminating unchecked misroutes

### Data Strategy

| Source | Usage |
|---|---|
| Call transcripts | Entity extraction and intent classification |
| Routing history | Context labels |
| Resolution outcomes | Validation and feedback signal |

Issue codes for common problems are confirmed to exist, providing clean classification targets.

### Success Metrics

- Reduction in 911 misroutes
- Improvement in routing accuracy over current baseline
- Reduction in cost per resolved ticket